

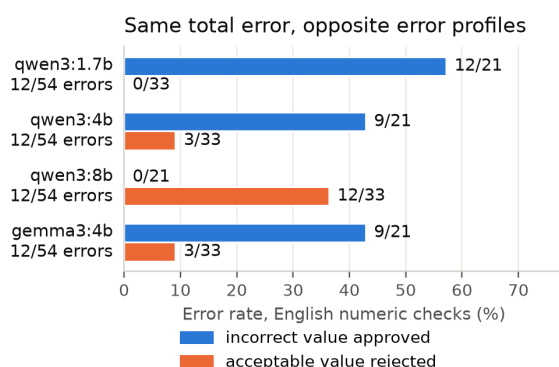
Who grades the grader?

APEX-Invariance: understanding what changes when APEX changes its judge

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APEX's scores depend on the judgment of the model grading the work. Improving that judge can involve a trade-off. In APEX-Agents 1.1, Mercor improved detection of scattergunned answers while false negatives rose from 5.3% to 8.0% relative to the naive grader [1]. That raises a practical question for the next upgrade: **which kinds of work will the new judge grade more reliably, and where will it introduce errors?**

I propose APEX-Invariance to help answer that question. It would complement Mercor's validation against human labels with controlled tests that explain when and why a verdict changes. By analyzing numeric, semantic and presentation criteria separately, the evaluation would help the APEX team identify which parts of an upgrade to adopt, which need further review, and which could benefit from a deterministic check.



I began investigating this problem by auditing 100 public APEX tasks and reproducing 19 numeric criteria from their source files. Across four local studies, I collected 2,394 valid judgments. The first comparison showed why total accuracy is not enough: four models each made 12 errors in 54 English numeric judgments, yet one model's errors were all false approvals and another's were all false rejections. A team choosing between them on accuracy alone would miss that difference. These local results do not establish failures in Mercor's judge, but they provide a working test suite and a specific question to bring to its reference configuration.

From an error count to an explanation

The method follows each criterion through changes whose correct effect is established in advance. If an answer expresses the same quantity in another numeric format, its verdict should stay the same; if that quantity moves outside the accepted tolerance, the verdict should reverse. In a follow-up local test, Qwen accepted -28 but rejected the equivalent -2.8E+1. For more complex work, experts would identify which criteria should change after a requirement is revised or an upstream calculation is corrupted. Matched unaffected responses and repeated judgments would distinguish a meaningful reaction from indiscriminate sensitivity or ordinary instability.

Building on behavioral judge testing [2] and selective acceptance [3], I would use these comparisons to test whether rubric dependencies capture how errors propagate and whether a numeric checker can reduce grading error while leaving ambiguous cases for review. The contribution would be measured against ordinary held-out validation: does the method reveal reproducible failures that accuracy misses, and does acting on those findings reduce grading error or review cost?

Building it with Mercor

I would start by aligning the suite with Mercor's reference judge and existing validation tools, then work with its experts on a pilot of up to five public task families within a 40-hour expert budget. Their independent judgments would establish the professional validity of the controls and the cost of extending them. If the pilot supports expansion, the six-month project would target 60 families, with separate calibration and test sets and all variants of a task kept together for analysis.

Mercor would receive an expert-reviewed dataset, a regression suite it could rerun at each judge upgrade, and a report connecting each criterion class's errors to an actionable recommendation. Automatic-grading recommendations would require enough independent evidence to meet agreed error limits; otherwise, the report would identify what remains uncertain and what further review is needed.

As first author of MeSHClass-ES and AnatEM-ES (BioNLP 2026), I released Spanish biomedical resources with reproducible translation and evaluation pipelines [4]. I would bring that experience and the completed local experiments to Mercor. A separately scoped extension could examine grading across languages and professional contexts, starting with Spanish and Latin America. Local experts would validate both translations and adaptations to local professional requirements, giving Mercor evidence to assess opportunities in new evaluation markets. I am based in Bogotá, available full time for six months, and willing to relocate if needed.

[1] APEX-Agents 1.1 · [2] Chen et al., arXiv 2608.24419 · [3] Badshah et al., arXiv 2608.17994 · [4] BioNLP 2026 · Data: APEX-v1-extended e0db951

For the full proposal, research hypotheses, references and reproducible experiment code, see github.com/SantiagoM99/mercors_research_fellowship.